

## Article

# Fake News Detection and Classification: A Comparative Study of Convolutional Neural Networks, Large Language Models, and Natural Language Processing Models

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**Abstract:** In an era where fake news detection has become a pressing issue due to its profound impacts on public opinion, democracy, and social trust, accurately identifying and classifying false information is a critical challenge. In this study, the effectiveness is investigated of advanced machine learning models—convolutional neural networks (CNNs), bidirectional encoder representations from transformers (BERT), and generative pre-trained transformers (GPTs)—for robust fake news classification. Each model brings unique strengths to the task, from CNNs’ pattern recognition capabilities to BERT and GPTs’ contextual understanding in the embedding space. Our results demonstrate that the fine-tuned GPT-4 Omni models achieve 98.6% accuracy, significantly outperforming traditional models like CNNs, which achieved only 58.6%. Notably, the smaller GPT-4o mini model performed comparably to its larger counterpart, highlighting the cost-effectiveness of smaller models for specialized tasks. These findings emphasize the importance of fine-tuning large language models (LLMs) to optimize the performance for complex tasks such as fake news classifier development, where capturing subtle contextual relationships in text is crucial. However, challenges such as computational costs and suboptimal outcomes in zero-shot classification persist, particularly when distinguishing fake content from legitimate information. By highlighting the practical application of fine-tuned LLMs and exploring the potential of few-shot learning for fake news detection, this research provides valuable insights for news organizations seeking to implement scalable and accurate solutions. Ultimately, this work contributes to fostering transparency and integrity in journalism through innovative AI-driven methods for fake news classification and automated fake news classifier systems.

**Keywords:** fake news classification; fake news detection; fake news classifier; misinformation; disinformation; convolutional neural networks (CNNs); bidirectional encoder representations from transformers (BERT); generative pre-trained transformers (GPTs); natural language processing (NLP); information integrity



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## 1. Introduction

The rapid proliferation of information in the digital age has transformed the way individuals consume news and interact with content. While access to diverse perspectives can enhance public discourse, it has also given rise to a troubling surge in misinformation and disinformation. Misinformation, defined as false or misleading information spread

without intent to deceive, and disinformation, which is deliberately misleading, pose significant threats to informed decision-making and democratic processes. The consequences of these phenomena are far-reaching, affecting public health, political stability, and societal cohesion [1].

As online platforms become the primary source of news for many, the challenge of distinguishing credible information from false narratives has intensified [2]. Traditional fact-checking methods often struggle to keep pace with the sheer volume and speed of modern information dissemination. Consequently, this has created a pressing need for automated, scalable solutions that can identify and mitigate the spread of false information efficiently.

#### Problem Statement

Despite advances in artificial intelligence (AI), accurately detecting fake news remains a significant challenge. Current approaches often fail to generalize across diverse linguistic contexts and topics, and the trade-offs between model accuracy, adaptability, and resource efficiency are not well understood. Additionally, the high computational costs of fine-tuning large language models (LLMs) present barriers to their widespread adoption. Addressing these challenges is critical for building reliable tools that enhance information integrity in the news ecosystem.

#### Research Goals and Significance

In this study, these challenges are addressed by investigating the performance of state-of-the-art deep neural network (DNN) models, including convolutional neural networks (CNNs), bidirectional encoder representations from transformers (BERT), and generative pre-trained transformers (GPT), in classifying fake news. The research aims to achieve the following:

1. Evaluate the effectiveness of advanced AI models in fake news detection, focusing on large language models (LLMs).
2. Compare the performance of these models before and after fine-tuning using few-shot learning techniques.
3. Examine the costs and trade-offs associated with fine-tuning LLMs and their implications for real-world applications.
4. Explore the transformative potential of LLMs to provide automated detection and actionable insights for the news industry.
5. Investigate the following critical questions that have yet to be thoroughly examined in previous studies:
  - Q1: are traditional NLP and CNN models or LLMs more accurate in fake news detection tasks?
  - Q2: among the GPT-4 Omni family, which model performs best prior to fine-tuning?
  - Q3: after fine-tuning with few-shot learning, which model in the GPT-4 Omni family demonstrates superior performance?
  - Q4: what is the significance of the costs associated with fine-tuning LLMs, and how do these costs impact performance in the news sector?
  - Q5: how can LLMs be effectively leveraged to assess fake news, and what transformative effects can they have on the news industry through automated detection and actionable insights?

By addressing these objectives, in this paper, we seek to contribute to the development of practical, robust tools for combating misinformation. The findings aim to empower journalists, policymakers, and the public with effective strategies and technologies to navigate an increasingly complex media landscape.

The structure of this paper is as follows:

- Section 2 presents a comprehensive review of the existing literature on fake news detection and classification, with a focus on the application of DNN models in the news sector.
- Section 3 outlines the methodologies employed in this study, detailing the fine-tuning of models using few-shot learning techniques to ensure transparency and reproducibility.
- Section 4 reports the predictive results of the analyzed models, highlighting their performance both before and after fine-tuning.
- Section 5 provides an in-depth discussion of the findings, extracting actionable insights and advancing the discourse on leveraging AI to combat misinformation effectively.

## 2. Literature Review

The exponential growth in social media has amplified both the creation and dissemination of information, making it easier for misleading or false information to proliferate widely and rapidly. This phenomenon has necessitated the development of robust methods to detect fake news and misinformation, as both customer satisfaction and user engagement are heavily influenced by the credibility of shared content [3]. These aspects are significantly taken into account nowadays by organizations that want to draw the public's attention to the content they deliver [4]. The researchers in this field have employed various approaches, predominantly focusing on the extraction and utilization of key features that distinguish false from legitimate content. These features range from linguistic and stylistic elements within the text to metadata and behavioral patterns observable in how content is shared.

Before our own investigation, it was imperative to examine all the research studies on the topic, highlighting their approaches, characteristics, and key contributions. The criteria for selecting these papers were relevant to the problem of fake news detection, the recency in capturing the latest trends, and the methodology used, with studies employing various algorithms to detect fake news. All these studies are briefly presented in Table 1 and thoroughly reviewed later, leading to the presentation of our own investigation, starting in Section 3 and beyond.

**Table 1.** Summary of key findings from selected papers on fake news detection.

| Paper                         | Objective                                                                                         | Approach                                                                                                             | Results                                                                                                 | Contribution to the Field                                                                                                                                                   |
|-------------------------------|---------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reis et al. (2019) [5]        | Investigate supervised learning techniques for fake news detection in social media contexts       | Feature extraction from news articles and social media posts, supervised learning                                    | Identified critical features and revealed effectiveness of various feature sets for fake news detection | Introduced a novel set of features and provided insights into the challenges of detecting false information, emphasizing practical applications.                            |
| Pérez-Rosas et al. (2018) [6] | Address the challenge of misleading information in accessible media with fake news classification | Developed two novel datasets for fake news classification, conducted linguistic analysis and comparative experiments | Automated methods outperformed manual identification in fake news classification                        | Demonstrated the advantages of computational tools over manual approaches in identifying fake news and highlighted linguistic differences between fake and legitimate news. |

Table 1. Cont.

| Paper                        | Objective                                                                                                       | Approach                                                                                                                       | Results                                                                                        | Contribution to the Field                                                                                                                                                                  |
|------------------------------|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Al Asaad et al. (2018) [7]   | Examine the implications of the “post-truth” era and propose a framework for detecting fake news                | Supervised learning with feature extraction using Bag-of-Words and TF-IDF                                                      | Linear classification with TF-IDF achieved highest accuracy, bigram models were less effective | Emphasized the importance of feature selection and classification strategies for effective fake news detection, providing insights into the “post-truth” era’s impact on misinformation.   |
| Thota et al. (2018) [8]      | Propose a deep learning approach to fake news classification, addressing the binary classification limitation   | Neural network architecture to predict stance between headlines and article bodies                                             | Achieved an accuracy of 94.21% and a 2.5% improvement over previous models                     | Focused on the need for automated systems and emphasized the improvement over existing models by predicting nuanced relationships between headlines and bodies.                            |
| Kaliyar et al. (2020) [9]    | Introduce FNDNet, a deep CNN for fake news detection                                                            | CNN-based model that automatically learns discriminative features through multiple hidden layers                               | Achieved an impressive accuracy of 98.36%, outperforming existing techniques                   | Demonstrated the potential of CNN-based models for fake news detection and highlighted the automatic feature learning process, marking a significant improvement over traditional methods. |
| Yang et al. (2018) [10]      | Explore fake news classification by integrating textual and visual information using the TI-CNN model           | Deep learning model combining both textual and visual information for fake news classification                                 | Achieved effective fake news detection using both explicit and latent feature extraction       | Introduced an innovative approach by incorporating both textual and visual information, improving the robustness and accuracy of fake news classification.                                 |
| Singhal et al. (2019) [11]   | Introduce SpotFake, a multi-modal framework for fake news detection leveraging both textual and visual features | Multi-modal framework using BERT for text feature extraction and VGG-19 for image feature extraction                           | Improved performance by 3.27% (Twitter) and 6.83% (Weibo) over state-of-the-art results        | Demonstrated the effectiveness of integrating both textual and visual features for fake news detection, surpassing existing techniques.                                                    |
| Devarajan et al. (2023) [12] | Propose an AI-assisted deep NLP-based approach for detecting fake news across social media platforms            | Incorporates social features and deep learning across four layers: publisher, social media networking, enabled edge, and cloud | Achieved 99.72% accuracy and 98.33% F1 score                                                   | Significantly outperformed existing methods, offering a comprehensive approach that integrates social media features with deep NLP for improved detection.                                 |
| Almarashy et al. (2023) [13] | Enhance accuracy in fake news classification by using a multi-feature classification model                      | Extracts global, spatial, and temporal features from text using TF-IDF, CNNs, and BiLSTM                                       | Demonstrated superiority over previous methods in classification accuracy                      | Highlighted the benefits of combining multiple feature extraction techniques (global, spatial, and temporal) for improved fake news detection.                                             |

Table 1. Cont.

| Paper                       | Objective                                                                                             | Approach                                                                                                                                                   | Results                                                                                                             | Contribution to the Field                                                                                                                                                     |
|-----------------------------|-------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Oshikawa et al. (2020) [14] | Provide a comprehensive survey on the intersection of NLP and machine learning in fake news detection | Review existing datasets, task formulations, and NLP solutions                                                                                             | Emphasized the need for practical detection models to improve effectiveness                                         | Advocated for more refined detection models, highlighting the challenges of fake news classification and the importance of automatic detection methods.                       |
| Mehta et al. (2024) [15]    | Focus on the efficacy of NLP and supervised learning in classifying fake news articles                | NLP-based feature extraction followed by supervised learning                                                                                               | Achieved high accuracy, precision, recall, and F1 score                                                             | Demonstrated robust performance with NLP techniques and supervised learning, revealing significant contributors to successful classification and providing valuable insights. |
| Madani et al. (2023) [16]   | Propose a two-phase model combining NLP and machine learning for fake news detection                  | Hybrid method with curriculum learning, k-nearest neighbor algorithm                                                                                       | Demonstrated superior performance compared to benchmark models                                                      | Showcased the potential of hybrid feature extraction and machine learning methods to enhance the performance of fake news detection models.                                   |
| Zhou et al. (2019) [17]     | Propose a network-based pattern-driven approach for fake news detection                               | Analyzed patterns of fake news propagation through social networks using social psychological theories, applying network-level analysis                    | Outperformed existing state-of-the-art techniques                                                                   | Enhanced feature engineering for fake news detection by focusing on social network patterns, improving explainability of detection models.                                    |
| Conroy et al. (2015) [18]   | Explore hybrid detection approaches combining linguistic cues with network analysis                   | Combined content-based analysis with network-based insights to identify deception in online news                                                           | Provided a robust hybrid framework for fake news classification                                                     | Demonstrated the effectiveness of integrating multiple methodologies (linguistic- and network-based) to improve fake news detection and combat misinformation.                |
| Kozik et al. (2024) [19]    | Survey state-of-the-art technologies for fake news detection                                          | Categorized veracity assessment methods into linguistic cue approaches and network analysis techniques. Proposed a hybrid approach combining both methods. | Advocated for a hybrid approach of linguistic cues and network-based behavioral data to improve fake news detection | Provided operational guidelines for developing effective fake news classifier systems and emphasized the evolving challenges in the online news publication landscape.        |

Table 1. Cont.

| Paper                         | Objective                                                                               | Approach                                                                                                                                                                                                                        | Results                                                                                                                                             | Contribution to the Field                                                                                                                                         |
|-------------------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Farhangian et al. (2024) [20] | Address challenges posed by the proliferation of social networks in fake news detection | Introduced an updated taxonomy based on feature types, detection perspectives, feature representation methods, and classification approaches. Conducted an empirical study on feature extraction and classification techniques. | Transformer-based approaches demonstrated superior performance; optimal feature extraction methods are dataset-dependent.                           | Emphasized the value of combining multiple feature representation methods and classification algorithms, particularly for improved generalization and efficiency. |
| Alghamdi et al. (2023) [21]   | Detect COVID-19 fake news using transformer-based models                                | Fine-tuning pre-trained transformer models (BERT, COVID-Twitter-BERT) with downstream CNN and BiGRU layers                                                                                                                      | Achieved a state-of-the-art F1 score of 98% with CT-BERT augmented with BiGRU                                                                       | Highlighted the effectiveness of fine-tuning transformer models and augmenting them with neural network layers for COVID-19 fake news detection.                  |
| Mahmud et al. (2023) [22]     | Address news authenticity issues with socio-political influences and biased news        | Proposed a novel framework integrating blockchain technology, smart contracts, and incremental machine learning                                                                                                                 | Achieved initial accuracies of 84.94% for training and 84.99% for testing, improving to 93.75% and 93.80% after nine rounds of incremental training | Introduced blockchain and incremental machine learning to assess news credibility, demonstrating the potential of decentralized platforms for news verification.  |
| Yang et al. (2019) [23]       | Introduce an unsupervised method for fake news detection                                | Generative model using a Bayesian network, treating news truths and user credibility as latent variables                                                                                                                        | Achieved notable improvements over existing unsupervised methods                                                                                    | Introduced a generative, unsupervised approach to fake news detection, utilizing user engagement data to infer authenticity without labeled data.                 |
| Liu et al. (2020) [24]        | Develop FNED for early fake news detection                                              | Deep neural network with feature extractor, position-aware attention mechanism, and multi-region mean-pooling                                                                                                                   | Achieved over 90% accuracy within five minutes of news propagation, outperforming baselines with only 10% labeled samples                           | Proposed FNED, a model designed for early-stage fake news detection, achieving high accuracy with limited labeled data.                                           |
| Wani et al. (2023) [25]       | Focus on toxic fake news classification for COVID-19 misinformation                     | Machine learning techniques (SVM, random forest) and transformer-based models (BERT) for toxicity analysis                                                                                                                      | Linear SVM achieved 92% accuracy, with high F1, F2, and F0.5 scores                                                                                 | Introduced a toxicity-oriented approach for distinguishing toxic fake news, suggesting its effectiveness for misinformation detection.                            |



Table 1. Cont.

| Paper                          | Objective                                                                  | Approach                                                                                             | Results                                                                                                                                                       | Contribution to the Field                                                                                                                                          |
|--------------------------------|----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Kapusta et al. (2024) [26]     | Examine text data augmentation techniques for fake news classification     | Synonym Replacement, Back Translation, and Reduction of Function Words (FWD) for corpus augmentation | Back Translation improved accuracy in SVM and Bernoulli Naive Bayes models, FWD improved Logistic Regression, original corpus performed best in Random Forest | Introduced data augmentation techniques that enhance the performance of word embeddings and classifiers in fake news detection.                                    |
| Raja et al. (2023) [27]        | Address fake news detection in Dravidian languages using transfer learning | Fine-tuning mBERT and XLM-R pretrained models with adaptive learning strategies                      | Achieved 93.31% accuracy on Dravidian fake news dataset, outperforming existing methods                                                                       | Proposed a transfer learning approach for fake news detection in low-resource languages, demonstrating effectiveness with adaptive fine-tuning.                    |
| Liu et al. (2024) [28]         | Develop few-shot fake news detection (FS-FND) framework using LLMs         | Dual-perspective Augmented Fake News Detection (DAFND) model with multiple modules                   | Effective in low-resource settings, improving fake news detection through the integration of multiple modules                                                 | Introduced a few-shot detection framework using large language models, focusing on low-resource scenarios and in-context learning for fake news detection.         |
| Mallick et al. (2023) [29]     | Develop a cooperative deep learning model for fake news detection          | Incorporated user feedback to assess news trust levels and ranked news accordingly                   | Achieved 98% accuracy for fake news detection, outperforming many existing models                                                                             | Proposed a cooperative deep learning approach with user feedback, which refines the model through continuous engagement to improve fake news detection.            |
| Shushkevich et al. (2023) [30] | Address multi-class fake news detection with a BERT-based approach         | Used SBERT, RoBERTa, mBERT, and ChatGPT-generated synthetic data for class balancing                 | Superior performance to existing methods using a multi-class classification framework with true, false, partially false, and other categories                 | Expanded the framework for fake news detection from binary to multi-class classification, improving detection outcomes using BERT-based models and synthetic data. |

### 2.1. Feature-Based Detection Approaches

Recent studies have increasingly focused on effective methods for detecting fake news, particularly in social media contexts. Reis et al. (2019) [5] investigated supervised learning techniques, emphasizing the extraction of features from news articles and social media posts. They introduced a novel set of features and evaluate the predictive performance of existing approaches, revealing critical insights into the effectiveness of various features in identifying false information. Their findings underscore practical applications while identifying challenges and opportunities in the field.

Building on this, Pérez-Rosas et al. (2018) [6] addressed the challenge of misleading information in accessible media by presenting two novel datasets designed for fake news classification across seven news domains. They detailed the collection, annotation, and validation processes and conducted exploratory analyses of linguistic differences between fake and legitimate news. Their comparative experiments demonstrate the advantages of auto-

mated methods over manual identification, highlighting the importance of computational tools for addressing misinformation.

Additionally, Al Asaad et al. (2018) [7] examined the implications of the “post-truth” era, where emotional appeals often overshadow objective facts, leading to misinformation. They proposed a machine learning framework that utilizes supervised learning for fake news detection, employing models such as Bag-of-Words and TF-IDF for feature extraction. Their experiments reveal that linear classification with TF-IDF yields the highest accuracy in content classification, while bigram frequency models perform less effectively. This work emphasizes the significance of feature selection and classification strategies in developing effective detection tools.

## 2.2. Deep Learning Techniques

In contrast to traditional approaches, recent advancements in deep learning have significantly enhanced the capability to detect fake news. Thota et al. (2018) [8] proposed a deep learning approach to fake news classification, underscoring the need for automated systems in light of the increasing prevalence of misinformation. They argued that existing models often treat the problem as a binary classification task, limiting their effectiveness in understanding the nuanced relationships between news articles and their veracity. To address this, the authors presented a neural network architecture designed to predict the stance between headlines and article bodies, achieving an accuracy of 94.21%—a 2.5% improvement over previous models.

Furthering this research, Kaliyar et al. (2020) [9] introduced FNDNet, a deep CNN specifically designed for fake news detection. Unlike traditional methods that rely on hand-crafted features, FNDNet automatically learns discriminative features through multiple hidden layers. Their model, trained and tested on benchmark datasets, achieved an impressive accuracy of 98.36%, demonstrating substantial improvements over existing techniques. This research emphasizes the potential of CNN-based models in enhancing fake news classification and broadening understanding in this domain.

Finally, Yang et al. (2018) [10] explored the fake news classification challenge with the TI-CNN model, which incorporated both textual and visual information. Recognizing the impact of fake news on public perception, especially during significant events like the 2016 U.S. presidential election, the authors identified useful explicit features from text and images. They also uncovered hidden patterns through latent feature extraction via multiple convolutional layers. By integrating explicit and latent features into a unified framework, TI-CNN shows promise in effectively identifying fake news across real-world datasets.

## 2.3. Multi-Modal and Hybrid Approaches

The surge in fake news on social media necessitates advanced detection systems that can analyze multiple content types. In response, Singhal et al. (2019) [11] introduced SpotFake, a multi-modal framework designed for effective fake news classification. Unlike existing systems that rely on additional subtasks (such as event discrimination), SpotFake addresses fake news detection directly by leveraging both textual and visual features. The authors utilized advanced language NLP models like BERT for text feature extraction and VGG-19 for image feature extraction. Their experiments on the Twitter and Weibo datasets demonstrate improved performance, surpassing state-of-the-art results by 3.27% and 6.83%, respectively.

Expanding on this concept, Devarajan et al. (2023) [12] proposed an AI-assisted deep NLP-based approach for detecting fake news from social media users. Recognizing the limitations of traditional content analysis methods, their model incorporated social features and operated across the following four layers: publisher, social media networking, enabled



edge, and cloud. The methodology encompasses data acquisition, information retrieval, NLP-based processing, and deep learning classification. Evaluating the model on datasets such as Buzzface, FakeNewsNet, and Twitter, they reported an impressive average accuracy of 99.72% and an F1 score of 98.33%, significantly outperforming existing techniques.

Furthermore, Almarashy et al. (2023) [13] tackled the challenge of fake news classification by enhancing accuracy through a multi-feature classification model. Their approach extracted global, spatial, and temporal features from text, which were then classified using a fast learning network (FLN). The model consisted of the following two phases: global features are obtained using TF-IDF, spatial features through a CNN, and temporal features via bi-directional long short-term memory (BiLSTM). Experiments conducted on two datasets, ISOT and FA-KES, demonstrate the model's superiority over previous methods, underscoring the effectiveness of combining diverse feature extraction techniques.

#### 2.4. NLP and Machine Learning

The intersection of NLP and machine learning is pivotal in addressing the challenges of fake news detection. Oshikawa et al. (2020) [14] provided a comprehensive survey of this domain, highlighting the critical need for automatic detection methods due to the rapid dissemination of misinformation on social media. They systematically reviewed existing datasets, task formulations, and NLP solutions, discussing their potential and limitations. The authors emphasized the distinction between fake news classification and other related tasks, advocating for more refined and practical detection models to enhance effectiveness in combating misinformation.

Complementing this perspective, Mehta et al. (2024) [15] focused on the efficacy of NLP and supervised learning in classifying fake news articles. Their study demonstrated the application of NLP techniques for feature extraction from textual data, followed by the training of a supervised learning model. Using a dataset of fake news articles, they evaluated model performance through metrics such as accuracy, precision, recall, and F1 score. Their results indicate that the approach achieves high accuracy and robustness in classification. Furthermore, feature importance analysis reveals significant contributors to successful classification, providing valuable insights for addressing fake news in online media.

In addition, Madani et al. (2023) [16] addressed the growing concern of fake news with a two-phase model that combines NLP and machine learning. The first phase involved extracting both new structural features and established key features from news samples. The second phase employed a hybrid method based on curriculum learning, integrating statistical data and a k-nearest neighbor algorithm to enhance the performance of deep learning models. Their findings demonstrated the model's superior capability in detecting fake news compared to benchmark models, underscoring the potential of combining innovative feature extraction and advanced machine learning techniques.

#### 2.5. Network-Based Detection Approaches

The rise in fake news has intensified the need for innovative detection methods. Zhou et al. (2019) [17] proposed a network-based pattern-driven approach to fake news detection that transcends traditional content analysis. Their study emphasized the importance of understanding how fake news propagates through social networks, focusing on the patterns of dissemination, the actors involved, and their interconnections. By applying social psychological theories, the authors presented empirical evidence of these patterns, which are analyzed at various network levels—including node, ego, triad, community, and overall network. This comprehensive approach not only enhances feature engineering for fake news detection but also improves the explainability of the detection

process. Experiments on real-world data indicate that their method outperforms existing state-of-the-art techniques.

In a complementary vein, Conroy et al. (2015) [18] explored hybrid detection approaches that combined linguistic cues with network analysis to tackle deception in online news. Their work highlighted the potential of integrating multiple methodologies to enhance the effectiveness of fake news classification strategies. By leveraging both content-based and network-based insights, these hybrid approaches offer a more robust framework for identifying and combating misinformation in digital spaces.

## 2.6. Meta-Analytic and Comparative Studies

Kozik et al. (2024) [19] conducted a comprehensive survey of state-of-the-art technologies for fake news detection, defining the task as categorizing news along a veracity continuum with a measure of certainty. The authors highlighted the challenges posed by the evolving landscape of online news publication, where traditional fact-checking methods struggle against a deluge of content. They categorized veracity assessment methods into two primary types, as follows: linguistic cue approaches, often enhanced by machine learning, and network analysis techniques. The paper advocated for a hybrid approach that merges linguistic cues with network-based behavioral data, offering a more nuanced understanding of the factors influencing news veracity. Additionally, the authors proposed operational guidelines to facilitate the development of effective fake news classifier systems.

Building on these insights, Farhangian et al. (2024) [20] addressed the challenges posed by the proliferation of social networks in combating fake news. Their paper revisited definitions of fake news and introduces an updated taxonomy based on four criteria: types of features used, detection perspectives, feature representation methods, and classification approaches. They conducted an extensive empirical study evaluating various feature extraction and classification techniques in terms of accuracy and computational cost. Their findings indicate that optimal feature extraction methods are dataset-dependent, with context-aware models, particularly transformer-based approaches, demonstrating superior performance. The study emphasizes the value of combining multiple feature representation methods and classification algorithms, including classical ones, for improved generalization and efficiency.

## 2.7. Specialized Detection Models

Alghamdi et al. (2023) [21] investigated the detection of COVID-19 fake news using transformer-based models, noting the surge of misinformation during the pandemic as a significant public health concern. The paper evaluated various machine learning algorithms and the effectiveness of fine-tuning pre-trained transformer models, such as BERT and COVID-Twitter-BERT (CT-BERT), for this purpose. By integrating downstream neural network structures, including CNN and BiGRU layers, with either frozen or unfrozen parameters, the authors conducted experiments on a real-world COVID-19 fake news dataset. Their findings reveal that augmenting CT-BERT with a BiGRU layer yields a state-of-the-art F1 score of 98%, underscoring the promise of advanced machine learning techniques in combating misinformation.

Similarly, Mahmud et al. (2023) [22] addressed concerns surrounding news authenticity in the context of socio-political influences and biased news dissemination. They proposed a novel evaluation framework for Bengali language news that integrates blockchain technology, smart contracts, and incremental machine learning. This framework combined machine classification with human expert opinion on a decentralized platform to assess news credibility. The NLP model undergoes continuous training, achieving initial accuracies of 84.94% for training and 84.99% for testing, which improve to 93.75% and 93.80%

after nine rounds of incremental training. Their simulation on the Ethereum test network demonstrates the successful implementation of this innovative system, highlighting the potential of leveraging blockchain for enhancing news verification processes.

### 2.8. Emerging Trends and Novel Techniques

In exploring novel approaches to fake news detection, Yang et al. (2019) [23] introduced an unsupervised method utilizing a generative model. Acknowledging the rapid dissemination of news on social media and the challenges posed by traditional supervised learning methods, which require extensive labeled datasets, this study employed a Bayesian network to treat news truths and user credibility as latent variables. By utilizing user engagement data to infer the authenticity of news without labeled data, the authors demonstrate a notable improvement over existing unsupervised methods across two datasets.

Building on this idea of early detection, Liu et al. (2020) [24] developed FNED, a deep neural network designed for the timely identification of fake news on social media. They address the challenge of limited early-stage data by proposing a model with the following three innovative components: (1) a feature extractor that combines user text responses and profiles, (2) a position-aware attention mechanism to prioritize significant user responses, and (3) a multi-region mean-pooling mechanism for effective feature aggregation. Their experiments show that FNED achieves over 90% accuracy within five minutes of news propagation, significantly outperforming state-of-the-art baselines while requiring only 10% labeled samples.

In a related effort, Wani et al. (2023) [25] focused on toxic fake news classification in the context of COVID-19 misinformation. Recognizing the detrimental effects of toxic fake news on society, they collected datasets from various social media platforms, labeling instances as toxic or nontoxic through toxicity analysis. The study employed both traditional machine learning techniques (such as linear SVM and random forest) and transformer-based methods (including BERT) for classification. Their findings reveal that the linear SVM method achieved an accuracy of 92% alongside impressive F1, F2, and F0.5 scores. This research indicates that their toxicity-oriented approach effectively distinguishes toxic fake news from non-toxic content, suggesting a promising direction for future investigations into combating misinformation.

### 2.9. Augmentation and Transfer Learning

Kapusta et al. (2024) [26] examined text data augmentation techniques for enhancing word embeddings in fake news classification. They highlighted that contemporary language models require large corpora for training to effectively capture semantic relationships. To address the limitations of existing corpora, the authors explored the following three data augmentation methods: synonym replacement, back translation, and reduction of function words (FWD). By applying these techniques, they generated diverse versions of a corpus used to train Word2Vec Skip-gram models. Their results showed significant statistical differences in classifier performance between augmented and original corpora, with back translation particularly enhancing accuracy in support vector and Bernoulli naive Bayes models. In contrast, FWD improved logistic regression, while the original corpus yielded superior results in random forest classification. Additionally, an intrinsic evaluation of lexical semantic relations indicated that the back translation corpus aligned more closely with established lexical resources, suggesting improvements in understanding specific semantic relationships.

In a different context, Raja et al. (2023) [27] focused on fake news detection in Dravidian languages using a transfer learning approach with adaptive fine-tuning. Acknowledging

the challenge posed by fake news, especially in low-resource languages, they introduced the Dravidian\_Fake dataset for fake news classification in Dravidian languages and created multilingual datasets by combining it with the English ISOT dataset. Their approach involved fine-tuning the mBERT and XLM-R pretrained transformer models using adaptive learning strategies. The classification model demonstrated an average accuracy of 93.31% on the Dravidian fake news dataset, outperforming existing methods and proving effective for sentence-level classification in resource-constrained environments.

Liu et al. (2024) [28] proposed a novel few-shot fake news detection (FS-FND) framework utilizing LLMs. This approach aimed to distinguish fake news from real news in low-resource scenarios, leveraging the prior knowledge and in-context learning capabilities of LLMs. They introduced the dual-perspective augmented fake news detection (DAFND) model, which consists of several modules, as follows: a detection module identifies keywords in the news, an investigation module retrieves relevant information, a judge module produces two prediction results, and a determination module integrates these results for the final classification. Their extensive experiments on publicly available datasets demonstrated the effectiveness of DAFND, particularly in low-resource settings, highlighting its potential to improve fake news detection.

#### 2.10. Cooperative and Feedback-Based Models

Mallick et al. (2023) [29] developed a cooperative deep learning model for fake news detection in online social networks. Recognizing the rapid spread of fake news—which often distorts facts for viral purposes and causes significant societal issues such as misinformation and misunderstanding—the authors highlighted the limitations of existing detection algorithms, particularly their lack of human engagement. To address this challenge, their proposed model incorporates user feedback to assess news trust levels, with news ranking based on these assessments.

In their framework, lower-ranked news articles undergo further language processing to verify their authenticity, while higher-ranked content is classified as genuine. The model employs a CNN to convert user feedback into rankings within the deep learning architecture. Additionally, negatively rated news articles are reintroduced into the system to refine and retrain the CNN model. The proposed approach achieved a remarkable accuracy rate of 98% for detecting fake news, outperforming many existing language processing-based models.

#### 2.11. Toxic News and Multiclass Classification

Shushkevich et al. (2023) [30] explored the challenges of fake news detection (FND) in a landscape marked by the easy creation and sharing of information. While traditional FND research often relies on binary classification focused on specific topics, this study expanded the framework to a multi-class classification approach that categorizes news articles into true, false, partially false, and other categories. The authors examined the performance of three BERT-based models—SBERT, RoBERTa, and mBERT—while also leveraging ChatGPT-generated synthetic data to enhance the class balance in the dataset. They implemented a two-step binary classification procedure to further improve detection outcomes. Focusing on the CheckThat! Lab dataset from CLEF-2022, the authors reported superior performance relative to existing methods.

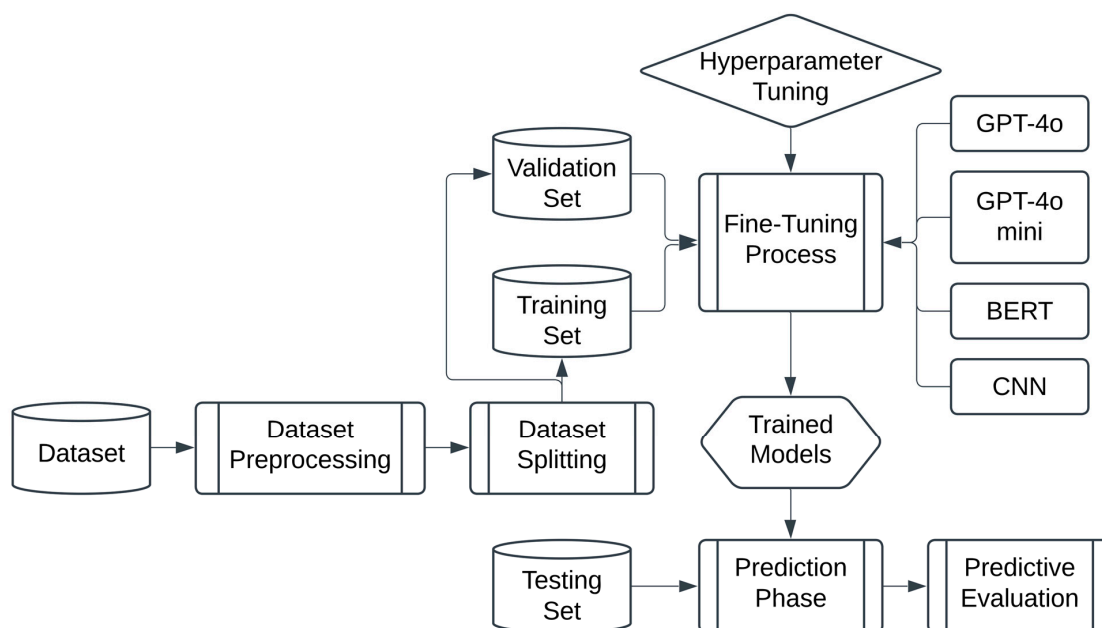
Based on the reviewed studies, the fight against fake news is a multifaceted challenge that requires continued innovation in detection methodologies. Recent advances in machine learning, deep learning, and user-centric approaches have laid a solid foundation for more effective detection systems.

### 3. Materials and Methods

In response to the growing challenges in detecting fake news across diverse platforms and domains, in our study, the aim was to advance current detection methodologies by leveraging state-of-the-art deep learning models. While existing models have demonstrated impressive accuracy, they often struggle with generalization to new types of misinformation, evolving content, and multimodal data. The performance of models can vary significantly depending on the dataset, context, and even the language in which the news is written. For instance, some models may perform well on certain platforms like Twitter or Weibo (as seen in Singhal et al. (2019) [11]), but their performance may degrade when applied to different domains, such as socio-political contexts (Mahmud et al. (2023) [22]). Even though high accuracy rates are reported in some studies, many models still struggle with data imbalance and the bias inherent in the datasets. Some models may overly rely on specific features (e.g., textual or visual), which may not generalize well across all types of fake news.

To address these challenges, we incorporated advanced techniques that offer both high performance and scalability. We chose to utilize LLMs such as GPT-4o and GPT-4o-mini, as they are pre-trained on large, diverse datasets and have demonstrated strong performance on tasks similar to fake news detection. These models possess a deep contextual understanding in the embedding space, making them highly effective for language tasks. Additionally, we employed CNNs to compare the performance of LLMs against traditional methods, providing valuable insights into their relative effectiveness for this specific problem. Finally, we used BERT as a cost-effective alternative to LLMs, as it can achieve strong results while requiring less computational power. To tailor each model to the unique requirements of fake news detection, we fine-tuned with few-shot learning all three approaches on our task-specific dataset, ensuring that they were optimized for this challenge. By comparing these models, we aimed to enhance the understanding of their relative strengths and weaknesses in detecting fake news across diverse platforms and contexts.

To provide a clearer explanation of the procedure followed in this study, we created a flowchart using the Lucidchart web app, presented in Figure 1.



**Figure 1.** Flowchart illustrating the dataset preprocessing, splitting, fine-tuning, predictions, and predictive evaluation process.



### 3.1. Dataset Cleaning, Preprocessing, and Splitting

To ensure the data quality and reliability of the dataset used in our study, we sourced the WELFake dataset from Kaggle, a reputable repository known for its wide range of datasets across different domains. The WELFake dataset consists of 72,134 news articles, with 35,028 labeled as real (0) and 37,106 labeled as fake (1). This dataset was compiled by merging four reputable sources—Kaggle, McIntire, Reuters, and BuzzFeed Political—making it comprehensive and representative of various topics. By utilizing this dataset, we reduced the risk of model overfitting, as its diversity provides a strong foundation for training robust machine learning models. The dataset is organized into the following four columns:

- Serial Number: the index of the article, starting at 0;
- Title: the headline of the article;
- Text: the full article content;
- Label: the classification label (0 for fake, 1 for real).

The total size of the dataset was 245.09 MB, with a usability rating of 10, ensuring its robustness and ease of use. It is available under the Attribution-NonCommercial 4.0 International license, allowing for public redistribution and adaptation for non-commercial purposes with proper credit [31,32].

#### 3.1.1. Dataset Preprocessing

To ensure the dataset was suitable for predictive modeling and fine-tuning, we employed a structured, multi-step preprocessing approach. Each phase of the preparation process was designed to maximize the data's relevance and effectiveness for our classification task.

1. Column Removal: We removed the "Unnamed: 0" column, which was deemed irrelevant to the analysis and redundant.
2. Empty Row Removal: We performed a thorough check for missing values across the "Title", "Text", and "Label" columns. Any rows containing missing values were removed to maintain the integrity of the data.
3. Column Merging: The "Title" and "Text" columns were combined into a new consolidated column, named "Text", to provide the model with a unified input that included both the article headline and content.
4. Label Standardization: The "label" column was standardized and renamed as "Label" for consistency across the dataset and to align with our modeling pipeline.
5. Text Length Restriction: We set a maximum length of 2560 characters for the "Text" column. This length was chosen to balance sufficient contextual information for training (particularly for CNN and BERT models) while maintaining memory and processing efficiency. After this truncation, the dataset contained 7573 entries labeled as 1 (real) and 7313 entries labeled as 0 (fake).
6. Data Standardization: Following the truncation, we standardized the dataset to ensure consistency and facilitate model convergence. After this step, we re-checked for any empty rows that might have been introduced and removed them, leaving a final dataset of 7568 real (1) and 7313 fake (0).
7. Balanced Sampling: To address potential class imbalances, we applied stratified sampling to select 5000 entries, with 2500 entries from each class (fake and real). This step ensured that the dataset was balanced, which is essential for training effective classification models.
8. ID Addition: A unique identifier (ID) was assigned to each entry to assist with tracking and error handling during the modeling process.

### 3.1.2. Dataset Splitting

We split the dataset into training, testing, and validation sets using a stratified approach within the `train_test_split` function to maintain the balance between the two classes. The split was carried out as follows:

- Training Set (80%): 3200 samples;
- Testing Set (20%): 1000 samples;
- Validation Set (20% of the training data): 800 samples.

This stratified splitting ensured that each set contained a balanced distribution of fake and real news, which is critical for model training. The training set was used for learning the underlying patterns and relationships within the data, while the validation set was used to fine-tune model hyperparameters and monitor performance. The test set was reserved to evaluate the final performance of the models after training.

### 3.2. LLM Prompt Engineering

Our objective was to create a prompt that seamlessly integrates with various LLMs, enhancing both the functionality and accessibility of their outputs through well-crafted code. We focused not only on crafting the prompt's content but also on making the output clear and easy to use, broadening its applicability across different use cases.

To achieve compatibility across multiple LLMs like GPT, Claude, and LLaMA—each with unique strengths and constraints—we first needed a detailed understanding of each model's characteristics. Designing a prompt that could elicit coherent and consistent responses from all these models, while maintaining ease of use, presented a complex challenge. To address this, we implemented the following two main prompt engineering strategies tailored to the specific needs of each model [33]:

- **Content Independent of Model Architecture:** We designed the prompt to be versatile and not dependent on any single model's framework. This flexibility ensured that it could be applied across different LLMs with minimal adjustment, focusing on clear communication of the task with relevant context and instructions interpretable by any LLM.
- **Structured Output for Accessibility:** Recognizing the importance of usability, we created a response format that aligned with coding and accessibility standards. The output was organized in compliance with the JSON standard, offering a logical, intuitive structure that meets both human readability and machine processing requirements.

After multiple testing rounds and refinements with different LLMs, we finalized a prompt that consistently produced outputs in the intended format, making it easy for both humans and models to interpret. The final version is illustrated in Listing 1.

#### Listing 1. Model-agnostic prompt.

```
conversation.append({'role': 'system',
'content': "You are an AI model tasked with predicting whether a news article is fake news. Respond with 0 for fake and 1 for not fake. Return your response in JSON format: {'fake': integer}."})
conversation.append({'role': 'user',
'content': f"Predict if the following news article is fake news (0 for fake and 1 for not fake). Please respond in JSON format like this example: {'fake': integer}. Please avoid providing additional explanations. Article text:\n{input['Text']}"})
```

### 3.3. Model Deployment, Fine-Tuning, and Predictive Evaluation

In this study, the purpose is to assess which of our four models most effectively identifies fake news content in the provided article, specifically for classification tasks. By pinpointing the model that best captures the context between words, we aim to create a robust tool for automatically extracting insights from news articles. This tool would empower individuals and organizations to make well-informed decisions, enhance business strategies, and drive improved results and overall success.

To accomplish this, each of the three models was tasked with generating predictions on the test set, both prior to and after fine-tuning, through few-shot learning. To ensure fair training conditions, we maintained consistent hyperparameters across all models, as follows: a learning rate of  $2 \times 10^{-5}$ , a batch size of 6, and three training epochs, using the Adam optimizer for fine-tuning.

The BERT and CNN models are typically designed to process inputs up to a 512-token limit, roughly equivalent to 2560 characters. For articles or texts exceeding this length, these models must truncate content to fit within the token constraint, which can lead to the loss of valuable context or critical information from longer texts. While techniques such as chunking (dividing texts into manageable segments) or hierarchical processing (analyzing segments in separate parts and then synthesizing results) are potential solutions for handling longer inputs, these methods add considerable complexity to the processing pipeline [34]. Specifically, these approaches can require additional layers of interpretation and synthesis to combine segmented outputs accurately, potentially affecting both the efficiency and precision of the results.

In contrast, models like GPT-4o—especially those configured with extended context windows—can handle significantly longer inputs than 512 tokens. Some versions of GPT-4 can process inputs up to 128,000 tokens [35], enabling the model to accommodate entire articles, books, or other complex documents in a single pass. This large token capacity allows GPT-4 to capture and integrate information across a much broader context, resulting in a more comprehensive understanding of lengthy texts without requiring chunking or hierarchical processing.

However, these significant differences in token capacity between BERT/CNN models and LLMs such as GPT-4 present challenges when trying to compare their performance directly. To enable a fair comparison, longer articles were removed from the training, validation, and test sets, as described in Section 3.1.1, ensuring that only articles with a maximum length of 2560 characters remained. This selection process helped standardize input sizes across models, limiting them to a shared input length, which allowed for a balanced assessment without the need for complex chunking methods or truncation inconsistencies.

Below, we present the deployment strategy for each model, outlining our tailored approach to applying them for fake news detection and classification tasks.

#### 3.3.1. GPT Model Deployment and Fine-Tuning

In this phase, we deployed models from the GPT Omni family, specifically the gpt-4o and gpt-4o-mini versions, both in their original (base) forms and after additional training (fine-tuning), to classify news articles as either fake or not fake. Initially, we used the base models in a zero-shot setting, applying the prompt in Listing 1 to make predictions on the test set without any prior fine-tuning. Due to their comprehensive pre-training on extensive datasets, these GPT models can provide relatively accurate predictions even without further training.

To enable interaction between our software and the GPT models, we utilized OpenAI's official API, which allowed us to submit prompts with article text from the feature column

(text) and receive predictions in JSON format. We saved the predictions from each model in separate columns within the test\_set.csv file for easy comparison.

During fine-tuning, these models were further fine-tuned to enhance their accuracy by learning from prompt–response pairs in our training dataset. This additional training enabled the models to better capture subtle patterns and intricacies within the data. We employed a multi-epoch training strategy, which allowed the models to improve iteratively across multiple passes through the data, resulting in more precise predictions and better overall task performance. This iterative approach was key to equipping the models with a nuanced understanding that enhanced their predictive accuracy and insights in the later prediction phase.

To conduct the fine-tuning, we created two JSONL files that included pairs of prompts and their corresponding completions, as outlined in Listing 2.

**Listing 2.** Prompt and completion pairs—JSONL files.

```
{
  "messages": [
    {
      "role": "system",
      "content": "You are an AI model tasked with predicting whether a news article is fake news."
    },
    {
      "role": "assistant",
      "content": "Respond with 0 for fake and 1 for not fake. Return your response in JSON format: {'fake': integer}."
    },
    {
      "role": "user",
      "content": "...",
    },
    {
      "role": "assistant",
      "content": "{'fake': 1}"
    }
  ]
}
```

Following the creation of the validation and training JSONL files, we carried out two fine-tuning tasks by uploading the JSONL files through OpenAI’s user interface. Each fine-tuning task was assigned a unique job ID for tracking, as follows: ftjob-9qj8DZgmx07iwoJsn8JOhx7c and ftjob-v10NfwXqreR6rmZNFuNOeqLV.

The first fine-tuning task was conducted with the gpt-4o model, which was trained on a dataset containing a total of 2,172,192 tokens. The model’s training loss started at 0.6997 and gradually decreased to 0.0021, indicating consistent improvement in fitting the training data. After completing the fine-tuning, the model’s validation loss—a measure of its ability to generalize—stood at 0.0037, demonstrating effective learning across the dataset.

The second task involved fine-tuning the gpt-4o-mini model on the same dataset with 2,172,192 tokens. This model began with an initial training loss of 0.9899, which also dropped to 0.0081 over the course of fine-tuning. Its final validation loss reached 0.0075, reflecting its ability to generalize successfully. These training metrics underscore both models’ suitability for the classification task and their robust learning during fine-tuning.

Once the fine-tuning process was complete, we used both models to generate predictions on the same test set they had initially seen in their base versions. The results from each model were then stored in individual columns within the dataset, allowing for an organized comparison between the fine-tuned and base model predictions.

As previously mentioned, the hyperparameters used during fine-tuning were consistent across all models: a learning rate of  $2 \times 10^{-5}$ , a batch size of 6, and three training epochs, ensuring fair training conditions. It is important to note that predictions and fine-tuning for the GPT models were conducted via the official OpenAI API, where we only had control over the batch size, learning rate, and the number of epochs. Furthermore, according to statements from Azure CTO Mark Russinovich [36], Azure leverages low-rank adaptation (LoRA), parameter-efficient fine-tuning (PEFT), and DeepSpeed tech-

niques to optimize GPU usage and improve memory efficiency during the fine-tuning of its GPT models.

### 3.3.2. BERT Model Deployment and Fine-Tuning

In this phase, we focused on training the BERT model, specifically the bert-base-uncased variant, to perform the same classification task that had been previously tackled by the LLMs [37]. For this, we used the BertForSequenceClassification class from the Hugging Face Transformers library. This variant of BERT incorporates an additional classification head specifically designed for sequence classification tasks. The classification head generally includes a fully connected layer, enabling BERT to convert its output into class probabilities, making it suitable for tasks like fake news detection and fake news classification.

The bert-base-uncased model architecture consists of 12 transformer layers, each with 768 hidden units and 12 attention heads, totaling around 110 million parameters. These self-attention mechanisms within BERT are highly effective at capturing contextual dependencies across tokens within an input sequence [38].

For the prediction phase, we used the pre-trained bert-base-uncased model, which was loaded directly using the from\_pretrained method, taking advantage of the model's robust pre-trained capabilities [39].

During fine-tuning, we adapted the BERT model for fake news classification by training it on labeled data containing labeled news articles, specifying each article as either fake or not fake. This training process was conducted on Google Colab, leveraging the processing power of a Tesla V100-SXM2-16GB GPU [40]. The dataset was stored on Google Drive, with preprocessing steps that included tokenization via BERT's tokenizer to prepare the text input in a format suitable for BERT.

Fine-tuning involved careful adjustments to hyperparameters, and we employed the adaptive moment estimation (Adam) optimizer for efficient optimization. Training was executed over three epochs, with progress tracked via the tqdm library to visualize real-time updates [41]. Within each epoch, backpropagation and optimization were applied to refine the model's weights, while validation was conducted using the same dataset previously used with the GPT-4o models, ensuring consistency in the evaluation process.

Our fine-tuning approach was methodical, tailored to meet the specific demands of the fake news classification task, and aimed at optimizing the model's performance for this purpose. After fine-tuning, the trained model generated predictions on the same test set used by the GPT-4o models, allowing for a direct comparison.

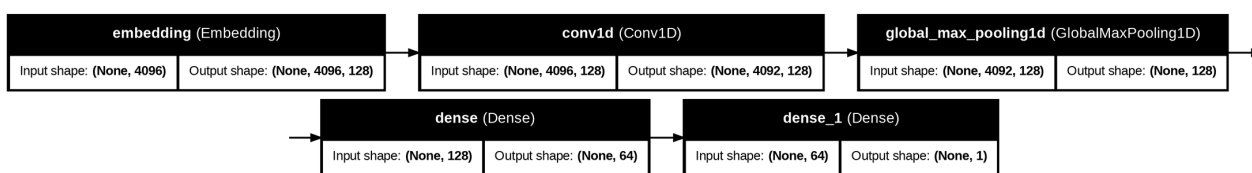
The full codebase, along with classes for training the BERT and GPT models, as well as metrics for training and validation loss and accuracy, is available in an ipynb notebook hosted on GitHub [42]. This resource also includes all relevant scripts and detailed documentation to facilitate replication and further experimentation.

### 3.3.3. CNN Model Deployment and Fine-Tuning

In this phase, we designed a CNN specifically aimed at classifying news articles into fake and non-fake categories. For data manipulation, we employed the Pandas library in Python, which facilitated the efficient handling and preprocessing of our datasets. The model development was carried out using TensorFlow's Keras API, a powerful tool for building and training deep learning models. To streamline our methodology, we created a custom class named CNNTraining, which encompassed essential functions such as hyperparameter initialization—setting the learning rate, number of epochs, batch size, and maximum sequence length—as well as mechanisms for storing training history and performance metrics.



The dataset was sourced from CSV files stored on Google Drive, enabling easy access and management. We employed Keras’s Tokenizer for the text tokenization process, which converted the articles into sequences of integers based on word frequency [43]. Following tokenization, we applied the `pad_sequences` function to standardize the input length across all samples, ensuring compatibility with the neural network. The architecture of our CNN included an embedding layer to transform word indices into dense vector representations, followed by a 1D Convolutional layer designed to extract local features from the text. This was complemented by a Global Max Pooling layer to reduce dimensionality and capture the most salient features, and two dense layers, one with a ReLU activation function for hidden representations and another with a sigmoid activation function for outputting probabilities of class membership (Figure 2).



**Figure 2.** Diagram of the CNN architecture used for fake news classification.

The model was compiled using the Adam optimizer, known for its efficiency in training deep learning models, and the binary cross-entropy loss function, which is suitable for binary classification tasks. The training process was executed over three epochs with predefined batch sizes, and validation metrics, such as loss and accuracy, were monitored throughout the training to assess model performance. Upon completion of the training phase, we assessed the model’s performance by utilizing the validation dataset, which allowed us to gauge its accuracy and effectiveness in classifying news articles. Following this evaluation, the trained model was then employed to make predictions on the test set, providing insights into its ability to generalize to unseen data. This two-step evaluation process ensured that we not only understood how well the model performed on the data on which it was trained, but also how reliably it could classify new articles, further validating its practical applicability in real-world scenarios.

## 4. Results

In Section 3, we undertook a comprehensive examination of the methodology used to implement and fine-tune the DNN models, thoroughly detailing the processes through which these models were fine-tuned to classify news articles in the test set as either fake or non-fake. In this section, a comparative analysis is provided of the four models, with a particular emphasis on their performance metrics prior to and following fine-tuning.

### 4.1. Overview of Fine-Tuning Metrics

During fine-tuning, we gathered key metrics for each model, including training loss, validation loss, training time, and training cost, all of which are detailed in Table 2.

Training loss measures the model’s effectiveness during the training phase by capturing the difference between its predictions and actual target values (e.g., 0 for fake and 1 for non-fake); a lower training loss suggests the model is learning effectively from the data.

Conversely, validation loss evaluates the model’s performance on a separate validation set that was not involved in training. This metric is essential for assessing the model’s generalization capability. Ideally, validation loss should decrease as the model improves, but if it starts to increase while the training loss continues to decline, this may indicate overfitting [44].

Table 2. Fine-tuning metrics.

| Model          | Resources             | Training Loss | Validation Loss | Training Time (Seconds) | Training Cost |
|----------------|-----------------------|---------------|-----------------|-------------------------|---------------|
| ft:gpt-4o      | API                   | 0.0021        | 0.0037          | 3353                    | USD 54.30     |
| ft:gpt-4o-mini | API                   | 0.0081        | 0.0075          | 1779                    | USD 6.52      |
| ft:bert-adam   | Tesla V100-SXM2-16 GB | 0.0294        | 0.0386          | 877                     | USD 2.54      |
| ft:cnn-adam    | Tesla V100-SXM2-16 GB | 0.6253        | 0.5884          | 47.90                   | USD 0.14      |

It is worth noting that direct comparisons of validation and training losses between fine-tuned CNNs, NLP models, and LLMs can be challenging due to differences in their architecture. However, comparisons among models with similar architectures, such as gpt-4o and gpt-4o-mini, are feasible due to shared structural characteristics and design principles. Since these models are variations within the same foundational framework, the differences observed in training and validation losses likely reflect the scale of the models rather than significant architectural differences. This shared foundation allows for more precise comparisons of their relative performance on similar tasks.

4.2. Model Evaluation Phase

Before we present our findings, it is crucial to highlight the importance of model evaluation. In the fields of machine learning and natural language processing, assessing models allows us to gauge their effectiveness, make data-driven decisions, and refine tuning to suit particular tasks. Table 3 provides a summary of each model’s evaluation, featuring essential metrics such as accuracy, recall, and F1-score.

Table 3. Comparison of model performance metrics.

| Model                        | Accuracy | Precision | Recall | F1     |
|------------------------------|----------|-----------|--------|--------|
| base:gpt-gpt-4o-2024-08-06   | 0.123    | 0.123     | 0.123  | 0.123  |
| base: gpt-4o-mini-2024-07-18 | 0.243    | 0.1969    | 0.243  | 0.2123 |
| ft:gpt-4o                    | 0.986    | 0.9861    | 0.986  | 0.986  |
| ft:gpt-4o-mini               | 0.986    | 0.9861    | 0.986  | 0.986  |
| ft:bert-adam                 | 0.975    | 0.9758    | 0.975  | 0.975  |
| ft:cnn_adam                  | 0.586    | 0.6334    | 0.586  | 0.5457 |

4.2.1. Pre-Fine-Tuning Evaluation

In the initial phase of our research, we deployed two baseline models from the GPT Omni family—gpt-4o and gpt-4o-mini—to assess their capabilities in zero-shot fake news detection and classification. The results, as summarized in Table 3, were unexpectedly low. The GPT-4o model achieved an accuracy of only 12.3%, while the smaller GPT-4o-mini outperformed it with an accuracy of 24.3%, nearly doubling the prediction accuracy of its larger counterpart. This unexpected performance gap raised questions about the models’ ability to handle this task without prior training. Initially, we hypothesized that the complexity of the prompt might have contributed to the models’ underperformance. To test this, we created two additional, simplified prompts intended to clarify the task for the models. However, even with these modifications, the classification results remained low, suggesting that the prompt complexity was not the primary issue.

This led us to reconsider our expectations for zero-shot capabilities in detecting fake news. In this task, distinguishing false news from legitimate information often requires subtle contextual understanding, which an untrained model might lack. This is particularly relevant given that even human readers sometimes struggle to identify fake news without prior knowledge of the topic. This reflection underscored a key limitation in the zero-shot

application of LLMs for nuanced classification tasks and pointed to the need for more targeted fine-tuning. Consequently, these findings provided a strong motivation to pursue further fine-tuning of the models, enabling them to develop the specific knowledge and context needed to excel in fake news detection tasks.

#### 4.2.2. Post Fine-Tuning Evaluation

The results following fine-tuning reveal a striking enhancement in model performance, with both fine-tuned gpt-4o (ft:gpt-4o) and its smaller variant, gpt-4o-mini (ft:gpt-4o-mini), achieving a remarkable accuracy of 98.6%. This represents an improvement of 86.3% and 74.3%, respectively, over their base models, underscoring the significant impact of task-specific tuning. While the GPT-4o model initially showed very low accuracy in zero-shot classification—less than half that of the mini counterpart—it overcame its challenges and demonstrated remarkable improvement, achieving the same 98.6% accuracy as the smaller model after fine-tuning. However, the identical accuracy of both models led to further investigation, as a more pronounced difference was expected due to their varying parameter sizes. To ensure the accuracy of these results, we conducted multiple rounds of model re-fine-tuning and verification using OpenAI's panel, each time confirming the same accuracy. This consistency suggests that the specific dataset and task may have enabled gpt-4o-mini to achieve a performance level equal to its larger counterpart, despite the fewer parameters.

These findings have compelling implications for model efficiency in targeted applications. The comparable performance of gpt-4o-mini suggests that for certain well-defined tasks, smaller models may serve as cost-effective alternatives to larger ones without significant sacrifices in accuracy. This parameter efficiency could benefit scenarios where computational resources or deployment costs are limited.

In contrast, the fine-tuned BERT (ft:bert-adam) and CNN (ft:cnn-adam) models produced notably different results. Ft:bert-adam achieved 97.5% accuracy, indicating a strong performance, while ft:bert-cnn lagged significantly with only 58.6% accuracy. This disparity highlights the critical role of model architecture and pretraining in the effectiveness of fine-tuning for specialized tasks. Transformer-based architectures like GPTs and BERT have an inherent advantage in NLP and classification tasks due to extensive pretraining on diverse, large-scale datasets. Conversely, CNN models, although highly effective for structured or image-based data, often require a large volume of labeled data to excel in more complex language-based tasks, where subtle semantic patterns are crucial for accurate classification.

The relatively low accuracy of the CNN model in this study may thus be due not to architectural limitations, but rather to an insufficient amount of labeled data to learn and recognize the intricate fake news patterns in the task. Unlike pre-trained transformers, which come with a broad foundational understanding from large, generalized datasets, CNNs must acquire such nuance from labeled data specific to the task. In applications like fake news categorization, where recognizing subtle semantic variations is essential, CNNs likely benefit from substantial labeled datasets to achieve high accuracy, highlighting the importance of dataset size and quality for such architectures.

## 5. Discussion

In the preceding sections, we explored the fake news detection and classification abilities of two LLMs from the same Omni family, alongside the BERT model and the CNN. To start, we assessed the base models in a zero-shot framework, where they predicted whether each news article in the test set was fake or non-fake purely based on its content, without any prior task-specific training. Following this, we conducted fine-tuning using

few-shot learning and efficient parameter tuning techniques, tailoring each model to a specialized training dataset. Once fine-tuned, the models were re-assessed on the same test set, and their results were analyzed in detail.

In this section, we will leverage these classification outcomes to address the research questions presented in the introduction.

### 5.1. Evaluating Traditional NLP Models vs. LLMs in Fake News Detection

- Research Question 1: are traditional NLP and CNN models or LLMs more accurate in fake news detection tasks?
- Research Statement 1: fine-tuned LLMs outperform traditional NLP and CNN models in fake news detection, achieving near-perfect accuracy.

In the comparative analysis of traditional NLP, CNN models, and LLMs in fake news identification tasks, in our study, several insights are underscored about model architecture, pretraining scope, and language comprehension capabilities. Fine-tuned LLMs, especially the GPT-4 Omni models, demonstrate a clear edge over both BERT-based models and CNN architectures, with accuracy levels reaching 98.6% for both GPT-4o and GPT-4o-mini. This near-perfect accuracy suggests that once fine-tuned, LLMs can spot fake news with high precision, making them valuable for tasks requiring fine semantic distinctions. The substantial margin over traditional models is particularly noteworthy because it showcases the advanced language understanding that LLMs bring to complex classification tasks, where subtle contextual cues determine the difference between legitimate content and false information.

- The role of pretraining and architecture: Unlike CNNs, which are primarily designed for pattern recognition in structured data like images [45], LLMs are built with transformer-based architectures that allow for deep attention mechanisms and sequence-based learning. These transformers, pretrained on vast and diverse datasets, are adept at capturing language patterns, idiomatic expressions, and subtle semantic relationships. In fake news detection, this translates to a model that can understand nuanced phrasing or stylistic cues typical of misinformation, even when these cues are subtle or context-dependent.
- CNN limitations: The CNN model (ft:cnn\_adam) in this study achieved only 58.6% accuracy, which is markedly lower than the transformer-based models. CNNs are effective at identifying repetitive, structured patterns but fall short when tasked with understanding the complexities of human language, especially when misleading content relies on nuanced or indirect language. Since CNNs do not inherently process sequential information as effectively as transformers, they struggle to recognize the sequential and contextual patterns often necessary for distinguishing fake news. Furthermore, CNNs require substantial labeled data tailored to the target task to perform well in NLP tasks, given their lack of extensive pretraining on varied textual data [46].
- Comparing BERT and GPT models in fake news classification: The BERT model (ft:bert-adam), while achieving a respectable 97.5% accuracy, still fell short of the fine-tuned GPT-4 Omni models. This difference, although minor, may be attributed to the GPT-4 Omni models' extensive pretraining and perhaps larger scale compared to BERT. Additionally, while both BERT and GPTs are transformer-based, GPT models are autoregressive, which means they are trained to predict the next word in a sequence, potentially enhancing their understanding of sentence flow and structure—elements that are crucial for detecting deceptive or misleading language. BERT's bidirectional nature gives it a slight advantage in understanding context but might limit its proficiency in tasks requiring generation or classification of nuanced language.

The findings suggest that transformer-based models, especially fine-tuned LLMs, are highly effective for fake news identification tasks. This has implications for organizations that require accurate and automated fake news classification, as LLMs can minimize false positives and false negatives more effectively than traditional models. The results also highlight that while BERT remains a viable option, GPT-based models offer a marginally higher level of performance, likely due to their extensive training on diverse language patterns. On the other hand, CNN models may not be suitable for fake news detection due to their architectural limitations and heavy reliance on task-specific labeled data.

### 5.2. Pre-Fine-Tuning Performance Assessment Within the GPT-4 Omni Family

- Research Question 2: among the GPT-4 Omni family, which model performs best prior to fine-tuning?
- Research Statement 2: prior to fine-tuning, GPT-4 Omni models perform poorly in fake news detection, highlighting the necessity for task-specific training.

Our analysis of the GPT-4 Omni family models before fine-tuning reveals several key insights about their performance capabilities, limitations, and the nature of zero-shot learning in specialized tasks like fraudulent news detection.

- Baseline performance and lack of task-specific knowledge: The low accuracy scores of 12.3% for GPT-4o and 24.3% for GPT-4o-mini underscore that both models lack the task-specific knowledge required for effective fake news detection in a zero-shot setting. These results suggest that while LLMs have extensive general language understanding, applying this to a nuanced, specialized task like misleading news categorization is challenging without specific tuning. Fake news classification often relies on recognizing subtle cues, phrasing patterns, and contextual red flags that are challenging for general-purpose models to identify without tailored training.
- The performance gap between the models—specifically, the nearly 50% difference in accuracy between GPT-4o and GPT-4o-mini—was unexpected. A plausible hypothesis is that GPT-4o, despite being larger and more powerful, may have been overfitted to its training data. This could cause the model to struggle in a zero-shot classification task if it overly relies on patterns from the training set that do not generalize well to new data. On the other hand, GPT-4o-mini, with its smaller parameter size, might have avoided overfitting, leading to better generalization in a zero-shot setting. It is a fact that sometimes smaller models can perform better in certain tasks because they learn to prioritize the most important features and avoid distractions from irrelevant data, while larger models might become bogged down by unnecessary complexity [47].
- Challenges of zero-shot fake news detection: Fake news detection is a complex task that requires not only general language understanding but also the ability to differentiate between legitimate and deceptive communication. Fraudulent content often imitates legitimate language, which makes it difficult to classify correctly without exposure to examples during training [40]. Zero-shot models, despite their general versatility, lack the fine-grained knowledge to identify these distinctions [48]. This is especially true in domains like fake news classification, where subtle stylistic or structural cues might signal fake news, and understanding these cues requires domain-specific data exposure.
- Implications of prompt complexity: Attempts to simplify prompts did not result in significant improvements in zero-shot performance, suggesting that prompt engineering alone may not be sufficient for bridging the knowledge gap in specialized tasks [49]. While prompt optimization can enhance zero-shot performance in some general tasks, its limited impact here implies that fake news detection requires more than refined prompting; it needs models that have been trained on data specific to the



task. This finding emphasizes that while LLMs are powerful, there are limits in what can be achieved through zero-shot learning alone in cases where the task requires deep contextual familiarity.

### 5.3. Fine-Tuning Impact on GPT-4 Omni Models with Few-Shot Learning

- Research Question 3: after fine-tuning with few-shot learning, which model in the GPT-4 Omni family demonstrates superior performance?
- Research Statement 3: after fine-tuning with few-shot learning, GPT-4o and GPT-4o-mini both achieve 98.6% accuracy, with GPT-4o-mini offering a resource-efficient alternative.

Examining the performance of GPT-4o and GPT-4o-mini after fine-tuning with few-shot learning offers insights into the effectiveness of fine-tuning and the strategic advantages of smaller models in targeted applications.

- High accuracy and comparable performance: Both GPT-4o and GPT-4o-mini achieved a remarkable accuracy of 98.6% after fine-tuning, suggesting that fine-tuning with few-shot learning equipped both models with a deep understanding of fake-related cues and patterns. This high accuracy indicates that fine-tuning enabled these models to internalize task-specific patterns, transforming general-purpose models into highly competent classifiers.
- Fine-tuning efficacy across model sizes: Fine-tuning proved equally effective for both the large and smaller models, suggesting that even a model with fewer parameters, like GPT-4o-mini, can achieve high accuracy when task-specific knowledge is provided through fine-tuning. This reinforces that model scaling is not always necessary for high performance in specialized tasks if effective fine-tuning methods, like few-shot learning, are applied. It also demonstrates that a smaller model, given the right training, can leverage its pre-existing language understanding to learn task-specific requirements efficiently.
- Scalability and flexibility in model deployment: The fact that GPT-4o-mini can achieve comparable performance to GPT-4o after fine-tuning suggests that smaller models in the GPT-4 Omni family can be scaled down without sacrificing substantial accuracy. This scalability is particularly beneficial for businesses or developers looking to deploy multiple models across various tasks, as smaller models require less computational power for deployment and can be trained more quickly [50]. Organizations that need to adapt quickly to new fake news detection patterns, for instance, might find GPT-4o-mini advantageous, as it combines high performance with adaptability and cost-effectiveness.
- Strategic model selection for application needs: For organizations with stringent accuracy standards in fake news detection, both models offer strong choices. However, GPT-4o-mini's identical performance to GPT-4o and its lower computational footprint make it particularly suitable for real-time fake news classifiers, mobile applications, or cloud deployments where resource limitations are a concern [51]. By achieving high accuracy with fewer resources, GPT-4o-mini serves as an example of how model selection can be aligned with specific operational and budgetary needs without compromising on task accuracy [52].

### 5.4. Cost-Performance Analysis of Fine-Tuning LLMs for Fake News Detection

- Research Question 4: what is the significance of the costs associated with fine-tuning LLMs, and how do these costs impact performance in the news sector?
- Research Statement 4: fine-tuning LLMs like GPT-4o incurs high costs, but GPT-4o-mini offers a nearly equal performance, making it a cost-effective and sustainable choice for the news sector.

The significance of fine-tuning costs for LLMs is particularly relevant in sectors like news, where budget constraints and scalability are critical. While fine-tuning improves model performance substantially, as seen with GPT-4o and GPT-4o-mini, the associated costs and computational resources required for larger models raise several strategic considerations for the news industry.

- **Cost-performance trade-offs:** Fine-tuning costs can vary dramatically between models, particularly as the model size and parameter count increase. While larger models like GPT-4o may offer accuracy improvements, these benefits often come with exponentially higher computational costs due to the additional resources needed for training and storage. The results of this study suggest that smaller models like GPT-4o-mini can achieve exactly the same accuracy (98.6%) as larger models, meaning that news organizations can achieve high performance without committing to the costs associated with the largest models.
- **Scalability and resource allocation in newsrooms:** Many newsrooms, especially smaller or independent ones, operate on limited budgets, making high-cost fine-tuning of large models unfeasible. GPT-4o-mini's near-parity in performance with GPT-4o after fine-tuning suggests that news organizations could allocate their resources more efficiently by selecting smaller models that require fewer computational resources. The cost of fine-tuning the mini model was only USD 6.52, compared to USD 54.30 for the larger model—a significant difference. This disparity was similarly large during the prediction phase. By using smaller models, organizations can implement robust AI solutions across multiple tasks—such as fake news detection, fake news analysis, and content moderation—without incurring prohibitive costs. This approach makes AI-powered solutions more scalable and accessible across diverse newsroom environments.
- **Sustainability and environmental impacts:** Computationally intensive fine-tuning contributes to energy consumption, which has significant environmental implications [53]. The use of a smaller model like GPT-4o-mini, which requires less power and computational time, aligns with sustainability goals by reducing the carbon footprint associated with model training. For news organizations committed to minimizing their environmental impact, smaller models represent a more sustainable alternative that still delivers a high performance. This consideration is becoming increasingly important for industries striving to balance technological advancement with environmental responsibility.

#### 5.5. Harnessing LLMs for Fake News Detection: Impact and Industry Transformation

- **Research Question 5:** how can LLMs be effectively leveraged to assess fake news, and what transformative effects can they have on the news industry through automated detection and actionable insights?
- **Research Statement 5:** LLMs can revolutionize fake news detection in the news industry by automating fact-checking, analyzing misinformation patterns, and optimizing journalistic workflows.

Leveraging LLMs like GPT-4o and GPT-4o-mini for fake news assessment offers a significant opportunity for the news industry due to their advanced language understanding and ability to detect subtle nuances in text. These models can automate the fact-checking process, help reduce misinformation, and ultimately enhance public trust in news outlets. Below, we outline the transformative impacts and specific advantages these models could offer to the news sector, along with potential opportunities for future research in this area.

- Automated fake news detection and verification: LLMs excel in detecting subtle linguistic cues, including tone, intent, and inconsistencies in phrasing that may indicate misinformation. By analyzing text with high sensitivity to such patterns, these models can flag potentially deceptive articles, posts, or statements [1]. Automating fake news detection enables near-instant identification of suspicious content, providing journalists and editors with a tool to screen and verify information before it reaches the public. This real-time verification can significantly reduce the spread of fake news by catching it early in the content distribution pipeline.
- Analyzing patterns and trends in misinformation: LLMs can analyze large datasets to identify recurring patterns in misinformation [54]. For instance, they can detect repeated themes, sources, or specific phrasing commonly associated with fake news, which helps newsrooms understand how misinformation is structured and spread. These insights allow media organizations to better understand the origins and propagation mechanisms of fake news, helping them create targeted counter-narratives and education campaigns to inform the public. Moreover, such analysis can assist journalists in investigating and debunking trends in misinformation at their root, reducing their overall impact.
- Efficient allocation of journalistic resources: Fake news detection traditionally requires extensive time and effort from journalists to verify sources, cross-check facts, and consult experts. With LLMs automating much of this initial verification process, journalists are free to focus on in-depth investigative reporting or nuanced storytelling. LLMs can serve as frontline tools, handling large volumes of content for preliminary screening and allowing human editors to prioritize the content that truly needs expert analysis [55]. This efficiency can lead to increased productivity in newsrooms, allowing them to cover more stories and provide richer, more balanced perspectives.
- Content moderation and community engagement: News outlets can deploy LLMs to moderate user-generated content, such as comments on articles or social media platforms, where misinformation often proliferates. By filtering out or flagging misleading comments in real-time, LLMs could enable news organizations to maintain respectful and informative discussions around their content. This content moderation creates a safer, more reliable environment for audience engagement, reducing misinformation on news platforms and fostering healthier community discourse [56].

## 6. Conclusions

In this study, we evaluated the effectiveness of LLMs, specifically the fine-tuned GPT-4 Omni models, in fake news detection for news content. Our results show that fine-tuned GPT-4o and GPT-4o-mini models achieved an impressive 98.6% accuracy, significantly outperforming traditional models like CNNs, which lagged at 58.6%. The GPT-4 models, despite their size difference, performed similarly post fine-tuning, highlighting the cost-effectiveness of smaller models without sacrificing accuracy. This research underscores the importance of fine-tuning for specialized tasks like fake news classification, where LLMs excel due to their ability to understand complex language patterns. It also emphasizes the potential for news organizations to leverage LLMs, particularly smaller models, to efficiently combat misinformation, balancing performance with computational cost. Ultimately, our findings contribute to the growing potential of AI in enhancing journalistic integrity and automating fake news detection, offering actionable insights for the future of news media.

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